

How Much of That Gain Is the Denominator? A Control for Legend Shrinking in Imbalanced Crop Mapping

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Abstract. Parcel-level crop maps are routinely evaluated with macro-averaged F1 over a catalog of classes the practitioner has already pruned, and the pruning is reported, when it is reported at all, as a property of the data rather than as a decision. We ask how much of the resulting gain belongs to the mechanism that pruned the catalog and how much to the smaller denominator that pruning creates. We formalize two operating mechanisms that trade coverage for macro quality, shrinking the promised legend and abstaining on low-confidence parcels, and evaluate them under a spatial-block protocol that chooses the operating point outside the data used to measure it, scores both mechanisms over the same class set, and never consults a label to decide what to deliver. Against a control that scores the untouched predictor over the same shrunken legend while delivering everything, most of the apparent gain disappears: on a first public benchmark the control accounts for most of it, on a second benchmark in a different year and region for almost all of it, and over the range where coverage stays complete for the entire gain, with every mechanism returning identical numbers. At matched coverage, abstention outperforms legend shrinking on both benchmarks, and retiring classes by low support, which is what practitioners report doing, is the weakest criterion we test and is not monotone in the number of classes retired. A macro-averaged figure computed over a pruned catalog is therefore uninterpretable without this control, which costs one line to compute.

Keywords: Crop mapping · Class imbalance · Selective classification · Evaluation protocol · Spatial cross-validation

1 Introduction

A parcel-level crop map promises a catalog. The catalog is a decision: someone chose which crop types the product will name and which it will not, and that choice is made long before any accuracy figure is computed. In practice the choice is rarely reported as a choice. It arrives in a sentence about how some classes had too few samples, and the evaluation that follows is computed over whatever survived.

This matters because the metric that dominates the literature on imbalanced crop mapping is macro-averaged F1, and macro-averaging divides by the number

of classes. Removing a class the model handled badly raises the macro average twice over: once by deleting a low per-class score from the numerator, and once by shrinking the denominator. The second effect requires no model, no mechanism, and no work. It is arithmetic. Yet a gain produced entirely by the second effect is indistinguishable, in a results table, from a gain produced by a method.

We take that ambiguity as the object of study. We consider two mechanisms that a practitioner can use to trade coverage for macro quality. The first shrinks the promised legend: the product names K of the C available classes and declines to answer wherever its unrestricted prediction falls outside them. The second keeps the full legend and abstains on the parcels where the predictor is least confident, in the tradition of selective classification [2,7]. Both reduce how much of the map is delivered; both raise the macro figure; and the question is how much of that rise either one is responsible for.

Our answer is a control, and it is almost embarrassingly cheap. Score the *untouched* predictor over the same shrunken legend while delivering everything. Whatever that control recovers was never bought by the mechanism. On a first public benchmark of eighteen agronomic classes, the control accounts for 88.3 % of the apparent gain from shrinking the catalog from eighteen classes to eight. On a second benchmark, in a different year, a different region and with a different model family, it accounts for 95.1 % at the pre-registered operating point. And over the range where coverage remains complete, it accounts for *all* of it: every mechanism we test returns the same number, to four decimals, while the macro figure climbs by 0.2447.

Two further results follow from the same protocol. At matched coverage, abstaining on low-confidence parcels outperforms shrinking the legend on both benchmarks, which is the opposite of what the practice we set out to formalize assumes. And retiring classes by low support, which is the criterion practitioners describe using, is the weakest of the three we test and is not monotone in the number of classes retired: retiring more can make the delivered map worse.

The contributions are as follows. We formalize the two mechanisms and give a spatial-block protocol that chooses the operating point outside the data used to measure it, scores both mechanisms over the same class set so that their numbers are comparable, and never consults a ground-truth label to decide which parcels to deliver. We quantify the decomposition of the apparent gain into denominator and mechanism on two public benchmarks under an unchanged protocol. We order the three retirement criteria with paired bootstrap intervals and a multiplicity correction. And we show that a trained stacking meta-learner performs the same retirement implicitly and undeclared, and that the difference between measuring it inside and outside the sample is enough to reverse the conclusion about whether the ensemble helps at all.

2 Related Work

We organize related work by what each line of research establishes and where it stops, because the gap this paper addresses lies in the space between two literatures that rarely meet.

Selective classification. The option to abstain is old and well characterized. Chow’s rule fixes the optimal error–reject tradeoff for a known posterior [2], and the modern treatment extends it to deep networks with learned or calibrated rejection [7,8,25], with recent surveys mapping the design space [10,9]. Two refinements matter here: the observation that a single risk–coverage curve hides which groups pay for the abstention [11], and the extension of reject curves to precision and recall rather than error alone [5,16]. What this literature does not treat is abstention by *class*: its unit of rejection is the sample, so a mechanism that declines to name an entire category falls outside its formalism, and the interaction between that mechanism and a macro-averaged metric is not analyzed.

Class structure in crop mapping. Work on crop mapping does manipulate the class catalog, but as a modeling device rather than as an operating point. Turkoglu et al. organize crop types into a hierarchy and let the model back off to a coarser level when the fine-grained decision is unreliable [22], which is the closest existing mechanism to legend shrinking; the evaluation, however, reports overall accuracy on a single split, so the effect on a macro average and its variability across space are not measured. Pan-European harmonization efforts confront the same problem at the level of the nomenclature itself [20,3], and fine-grained mapping work notes the cost of rare classes without isolating it [12,13,4]. Across this literature the pruned catalog is reported as a property of the data.

Benchmarks and temporal models. We evaluate on two public parcel-level benchmarks. PASTIS provides Sentinel-2 time series with panoptic annotations over France [6]; BreizhCrops provides a larger parcel-level collection over Brittany with a native partition by administrative region and an extreme class distribution [18]. The temporal architectures we use as predictors are standard: temporal convolutions [14], self-attention over raw series [19], and vision transformers adapted to satellite image time series [21]. None of these are contributions of this paper; they supply the posteriors over which the protocol operates.

Evaluation protocol. That the protocol is spatial rather than random is not optional. Cross-validation that ignores spatial autocorrelation inflates estimates of predictive performance, sometimes severely [17,15]. We therefore choose every operating point on spatial blocks disjoint from the block it is evaluated on. The ensembling literature supplies the remaining ingredient, the stacking meta-learner whose implicit behavior we examine [23,24,1], and it is precisely Wolpert’s requirement of leakage-free partitions for training the second-level model that our in-sample versus out-of-sample comparison puts to the test.

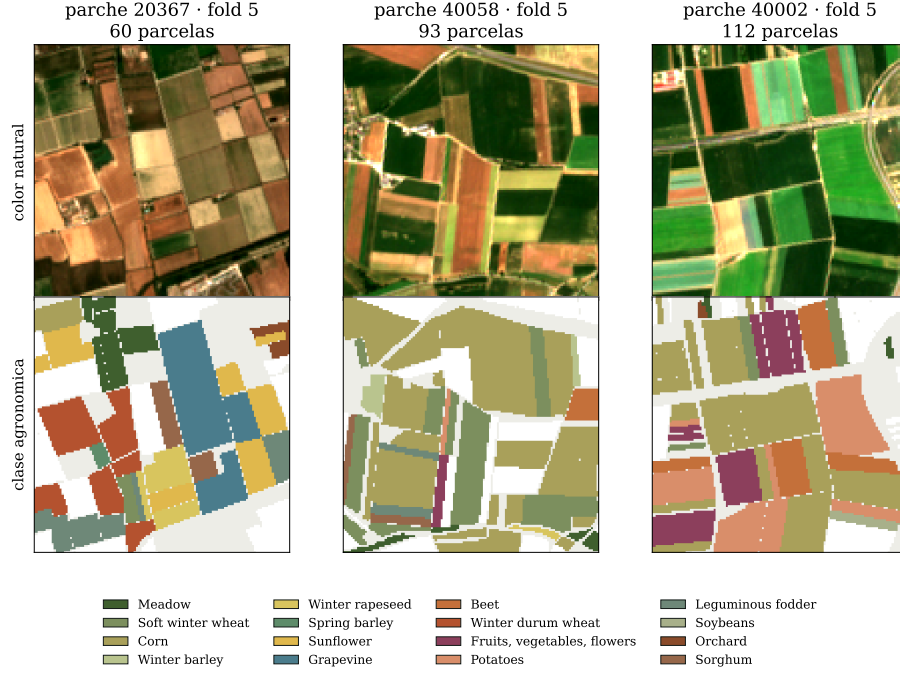


Fig. 1. Three patches of the held-out fold of the primary benchmark. Top: natural-color Sentinel-2 composite from the acquisition nearest to 1 July. Bottom: the parcel-level agronomic annotation. The unit of the product is the parcel, and the catalog this paper studies is the list of names the bottom row is allowed to use.

3 Materials and Method

3.1 Data

We use two public parcel-level benchmarks and never tune the protocol between them.

The **primary benchmark** is PASTIS-R [6], from which we take the held-out fold and aggregate the dense annotations to parcel level, yielding 16 640 parcels across 496 image patches and eighteen agronomic classes. Its imbalance is severe: the majority class holds 6 128 parcels and the minority class 103, a ratio of roughly 60:1.

The **replication benchmark** is BreizhCrops [18], Sentinel-2 L2A series over Brittany for the 2017 season, with nine agronomic classes and a native partition into administrative regions. We use regions `frh01` and `frh04` with a proportional stratified subsample of 30 000 parcels each, fixed seed, for a total of 60 000 parcels. Its imbalance is an order of magnitude worse than the primary benchmark: in the full index the majority class outnumbers the minority by up to 19 207:1, and in our subsample three of the nine classes hold two, seven and 289 parcels

Table 1. The two benchmarks. Both are described by the same 185-dimensional temporal feature vector, so the replication varies data, year, region and model family, but not representation.

	primary (PASTIS-R)	replication (BreizhCrops)
Season	2018–2019	2017
Region	metropolitan France	Brittany
Parcels scored	16 640	60 000
Classes	18	9
Majority class	6 128	18 113
Minority class	103	2
Imbalance ratio	60:1	9 057:1
Blocks	5 spatial (H3, 1 km buffer)	2 administrative regions
Predictor	pre-trained members	gradient boosting, leave-one-region-out

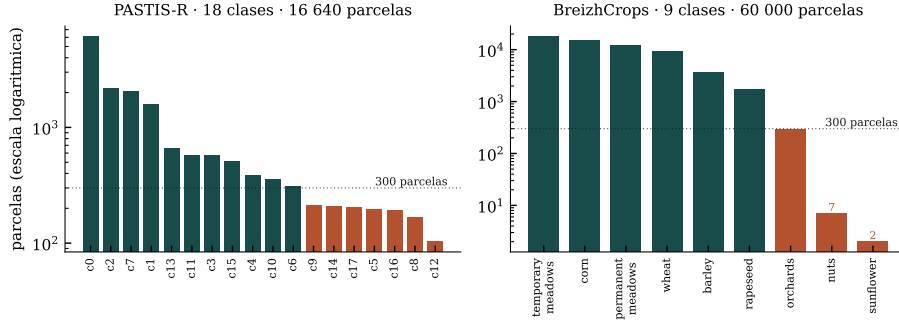


Fig. 2. Class support on a logarithmic scale. Bars below 300 parcels are drawn in the accent color: these are the classes every retirement criterion removes first. The replication benchmark’s tail is not merely thinner, it is degenerate — two and seven parcels — which is what makes the corridor of Sect. 4.1 possible.

respectively. Both benchmarks are described by the same 185-dimensional temporal feature vector, derived from seventeen spectral indices, so that the two experiments differ in data and model but not in representation.

3.2 Predictors

The protocol operates on posteriors and is agnostic to how they were produced. On the primary benchmark we take the two strongest of ten available members, a vision transformer for satellite image time series [21] and a gradient-boosted model over learned embeddings. On the replication benchmark we train a gradient-boosted classifier *leaving one region out*: each parcel’s posterior comes from a model that never saw its region, which makes every posterior we score a genuine out-of-region prediction. That classifier reaches macro-F1 0.497 9 and accuracy 0.742 1 on `frh01`, and 0.481 1 and 0.725 7 on `frh04`.

3.3 Blocks

Every operating point is chosen on data disjoint from the data that measures it. On the primary benchmark the blocks are spatial: an H3 tessellation at resolution five, clustered and separated by a 1 km buffer, giving five blocks. On the replication benchmark the index publishes no per-parcel coordinates, so the finest spatial structure available is the administrative region; we use the two regions as the two blocks. This is a real limitation and we treat it as one: with two blocks the resampling captures variation *within* regions and not *between* them, so we report the per-block difference for every contrast rather than a pooled interval alone.

3.4 The two mechanisms, and a third criterion

Let $\mathcal{C}$ be the full set of C classes, $p(\cdot | x)$ the predictor’s posterior for parcel x , and $\hat{y}(x) = \arg \max_{c \in \mathcal{C}} p(c | x)$ its unrestricted prediction.

Legend shrinking selects a subset $\mathcal{L} \subseteq \mathcal{C}$ with $|\mathcal{L}| = K$ on the training blocks, and on the evaluation block delivers a parcel when $\hat{y}(x) \in \mathcal{L}$, emitting $\arg \max_{c \in \mathcal{L}} p(c | x)$. We test two criteria for choosing $\mathcal{L}$: by per-class F1 on the training blocks, and by class support on the training blocks, the latter because it is what practitioners report doing.

Confidence rejection keeps $\mathcal{L} = \mathcal{C}$ and delivers the parcels whose $\max_c p(c | x)$ falls above a threshold. The threshold is the quantile of the block’s own confidences that matches the number of parcels the comparison mechanism delivered in that same block, so the two are compared at equal coverage.

Neither mechanism reads a ground-truth label to decide what to deliver. This is worth stating explicitly because the natural implementation of legend shrinking does read one: delivering the parcels whose *true* label lies in $\mathcal{L}$ is an oracle, and it inflates the mechanism’s apparent quality. Our delivery rule uses the unrestricted argmax, which is observable at inference time.

3.5 The aligned estimand

Two mechanisms that promise different catalogs cannot be compared by each one’s own macro average, because the averages are taken over different denominators. We therefore score every mechanism with

$$\text{F1}_{\text{aligned}} = \frac{1}{|\mathcal{L} \cap \mathcal{Y}|} \sum_{c \in \mathcal{L} \cap \mathcal{Y}} \text{F1}_c, \quad (1)$$

where $\mathcal{L}$ is the legend under test and $\mathcal{Y}$ the classes actually present among the delivered parcels. Both mechanisms in a contrast are scored over the *same* $\mathcal{L}$. We also report each mechanism’s native macro over what it individually promises, but only as a product-facing view; it is not comparable across mechanisms and we never use it for a contrast.

3.6 The control

The control that gives this paper its title is one line. For each legend $\mathcal{L}$ and each block, score the untouched predictor over $\mathcal{L}$ using Eq. 1 while delivering *every* parcel. No class is retired from the emission, no parcel is refused; only the denominator changes. If this control matches or exceeds a mechanism, the quality was bought by the choice of class set and not by the mechanism.

3.7 Inference

Contrasts use a paired bootstrap: one resample index is drawn per block and *both* mechanisms are rescored on it, so the two share the sampling noise. We draw 1 000 resamples and report the percentile interval of the difference of mean block-level aligned F1, together with the per-block differences. On the primary benchmark we additionally resample by image patch rather than by parcel, since parcels within a patch share an image and are not independent. The pre-registered principal criterion is $K = \lceil C/2 \rceil$; the remaining values of K form an exploratory family, corrected by the Holm–Bonferroni step-down procedure within each predictor and each benchmark.

4 Results

4.1 The corridor where every mechanism is identical

On the replication benchmark, shrinking the catalog from nine classes to six raises aligned macro F1 from 0.489 5 to 0.734 2, an apparent gain of 0.244 7. Over that entire range all four series — retirement by F1, retirement by support, confidence rejection, and the control that applies no mechanism at all — return the *same* value, and coverage stays at 1.000.

The reason is arithmetic and it is what makes the observation conclusive. The three classes retired first hold two, seven and 289 parcels out of 60 000; the unrestricted argmax essentially never predicts them, so removing them from the legend removes no parcel from the delivered map. No mechanism does anything. The metric rises by almost a quarter of a point on its own.

4.2 The decomposition transports

The same protocol, unchanged, gives the same qualitative answer on the primary benchmark. Shrinking from eighteen classes to eight raises aligned macro F1 from 0.561 2 to 0.805 2, an apparent gain of 0.244 0, of which the control recovers 0.215 5, or 88.3 %. At the pre-registered operating point of $K = 9$ the control recovers 87.1 % and the result holds for both of the two predictors we tested.

Two benchmarks, two years, two regions, two model families and one protocol therefore agree: the denominator dominates. Where they differ is in degree, and in the direction that matters for a practitioner: the replication benchmark, whose imbalance is an order of magnitude worse, leaves the mechanism even less to explain.

Table 2. Decomposition of the apparent gain on the replication benchmark, nine-class universe. “Denominator” is the control of Sect. 3.6: the untouched predictor scored over the same legend, delivering everything. $K = 5$ is the pre-registered principal criterion.

K	aligned F1	apparent gain	denominator	mechanism	coverage
9	0.4895	—	—	—	1.000
8	0.5507	+0.0612	+0.0612 (100.0%)	+0.0000 (0.0%)	1.000
7	0.6293	+0.1398	+0.1398 (100.0%)	+0.0000 (0.0%)	1.000
6	0.7342	+0.2447	+0.2447 (100.0%)	+0.0000 (0.0%)	1.000
5	0.8055	+0.3160	+0.3004 (95.1%)	+0.0156 (4.9%)	0.859
4	0.8579	+0.3684	+0.3352 (91.0%)	+0.0333 (9.0%)	0.806
3	0.9126	+0.4232	+0.3826 (90.4%)	+0.0406 (9.6%)	0.444

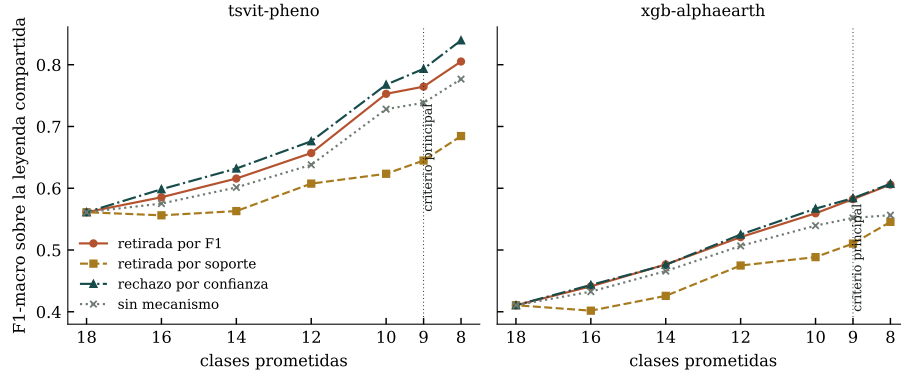


Fig. 3. The same frontier on the primary benchmark, for its two strongest predictors. The control that applies no mechanism (crosses) already recovers most of the distance travelled by the mechanisms, and retirement by support (squares) again trails the other two throughout.

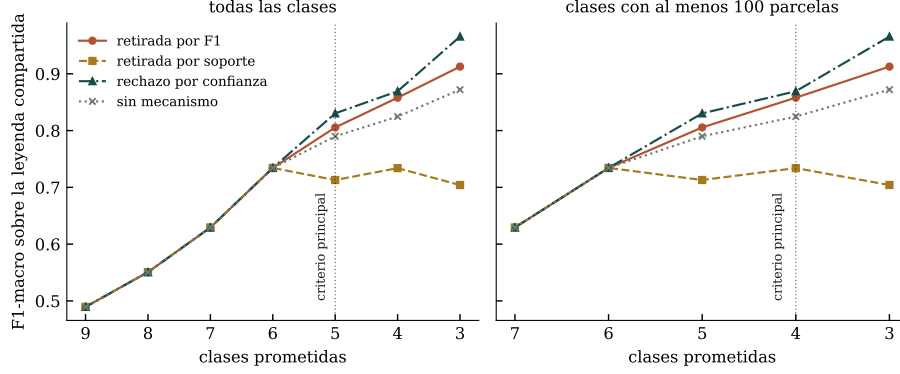


Fig. 4. Quality-coverage frontier on the replication benchmark, in the two pre-registered class universes. The four series are indistinguishable until $K = 6$ because coverage is complete and no mechanism acts; the dotted line marks the principal criterion. Retirement by support (squares) is the only series that falls as more classes are retired.

4.3 At equal coverage, abstention wins

Once coverage falls below one and the mechanisms begin to differ, they can be ordered. At the principal criterion of the replication benchmark, confidence rejection reaches 0.8303 against 0.8055 for legend shrinking at the same coverage, a difference of -0.0248 with a paired 95 % interval of $(-0.0272, -0.0225)$ and a Holm-adjusted p below 0.0001. Both regions agree in sign and magnitude, at -0.0248 and -0.0249 .

On the primary benchmark the difference has the same sign but is not distinguishable from zero: -0.0288 with an interval of $(-0.0414, +0.0080)$ and a Holm-adjusted p of 0.336. The honest summary across both benchmarks is therefore that we find no evidence that shrinking the legend outperforms abstention, and on the larger and more imbalanced benchmark we find significant evidence of the reverse.

4.4 Retirement by support is the weakest criterion, and it is not monotone

Retiring classes by low support is the criterion practitioners describe using, and it is the worst of the three on both benchmarks. On the replication benchmark it reaches 0.7129 at the principal criterion against 0.8055 for retirement by F1, a gap of 0.0926, which widens to 0.2085 at $K = 3$. On the primary benchmark the corresponding gap is between 0.12 and 0.13.

It also does something neither other criterion does: it *degrades* as more classes are retired, from 0.7129 at $K = 5$ to 0.7041 at $K = 3$. Inspecting the legends explains why. Support-based retirement removes rapeseed, which holds 1718

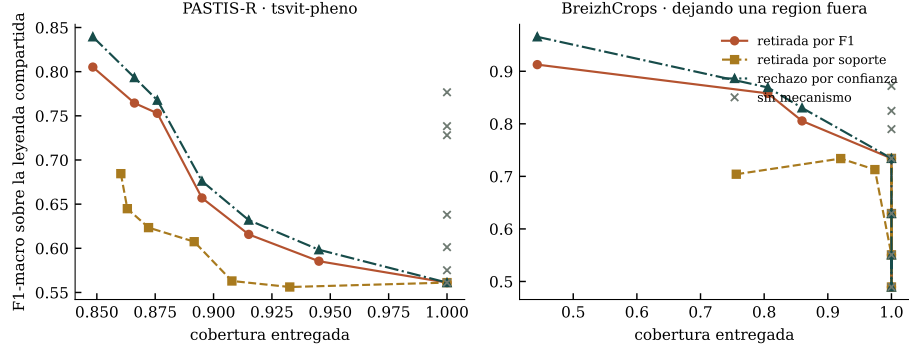


Fig. 5. The same measurements plotted against delivered coverage rather than legend size, which is how the selective classification literature reads them. The control (crosses) sits on the coverage-one vertical by construction, since it never withholds anything; the point of the figure is how many of those crosses lie *above* the mechanisms’ curves at lower coverage.

parcels and which the model separates well, while retaining both meadow classes, which the model confuses with each other. The criterion optimizes sample size, which is not the same quantity as the quality of what is delivered. Fig. 6 shows the substitution directly.

4.5 A trained meta-learner retires classes implicitly

The mechanism this paper formalizes is already being applied, undeclared, by trained ensembles. On the primary benchmark, a stacking meta-learner over three members drives one class of 355 parcels to an F1 of exactly 0.0000, while the best single model reaches 0.7790 on that same class. The meta-learner has retired a class from the product without anyone choosing to, and without it appearing anywhere in a results table.

The regime in which such a meta-learner is measured is decisive. Scored in-sample — refitted on all parcels after cross-validation, which is the default of many stacking implementations — it reports 0.7470. Scored leak-free by pooling its spatial out-of-fold posteriors and scoring once, it reports 0.6789. Averaging per-block macro scores instead of pooling reports 0.5360. The leakage accounts for 0.068 and the choice of aggregation for a further 0.143; conflating the two leads to reporting 0.536 as the honest figure when it is 0.679, penalizing the model for a measurement artifact on top of its actual error. Against the best single model, the leak-free meta-learner loses by 0.0572 with an interval of $(-0.0644, -0.0499)$ that excludes zero.

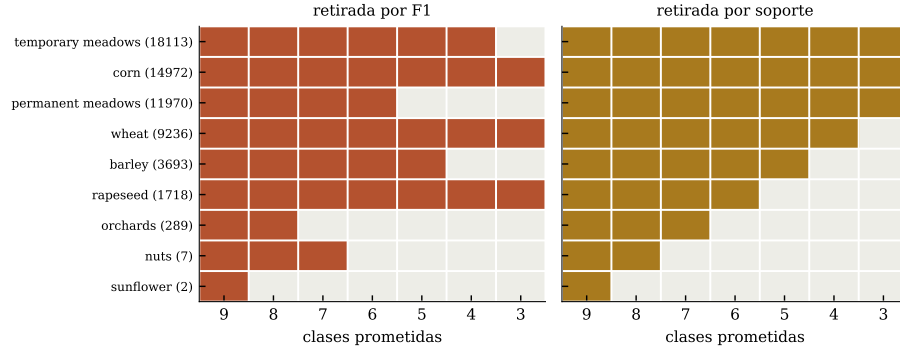


Fig. 6. Which classes each criterion keeps as the legend shrinks, on the replication benchmark; a filled cell means the class is promised. Support-based retirement (right) drops rapeseed at $K = 5$ while keeping both meadow classes, which the model confuses with one another; quality-based retirement (left) does the reverse. The two criteria disagree about exactly the classes that decide the outcome.

Table 3. Per-class F1 on the primary benchmark, leak-free, ordered by support. The trained arbiter is the only combiner that zeroes a class, and it does so to one of 355 parcels that every other combiner — including the untrained mean — handles above 0.77. The retirement is therefore a learned behavior, not a property of combining.

Class	support best single		mean weighted vote trained arbiter		
Meadow	6 128	0.9099	0.8980	0.9046	0.9138
Corn	2 196	0.9295	0.8683	0.9053	0.9274
Grapevine	2 067	0.9066	0.9141	0.9154	0.9124
Soft winter wheat	1 584	0.8922	0.8466	0.8712	0.8886
Leguminous fodder	660	0.5087	0.3498	0.4707	0.5243
Winter rapeseed	387	0.9358	0.9153	0.9275	0.9377
Winter durum wheat	355	0.7790	0.7752	0.7782	0.0000
Spring barley	198	0.6667	0.4691	0.5879	0.5228
Mixed cereal	193	0.4949	0.3525	0.4302	0.2709
Potatoes	103	0.5595	0.5143	0.5638	0.5823

Table 4. The same stacking meta-learner under three estimators. These are not three plausible choices: the first is contaminated and the third is penalized by an aggregation artifact.

Estimator	macro F1 what it measures
Refit on all parcels	0.7470 in-sample for the meta-learner
Pooled spatial out-of-fold	0.6789 leak-free, scored once
Mean of per-block macros	0.5360 leak-free, penalized by aggregation

5 Discussion

What each mechanism actually buys. Legend shrinking and confidence rejection both trade coverage for macro quality, but they charge different customers. Shrinking the legend refuses an entire crop type everywhere: a farmer growing it receives nothing, in every parcel, in every region. Abstention refuses individual parcels scattered across all crop types, so no grower is systematically excluded, which is the selective classification literature’s own concern about who pays for the abstention [11]. Our measurements say that abstention is also the better buy at equal coverage. The two arguments point the same way, and we know of no operational setting in which the reverse is preferable except one: when the downstream consumer requires a fixed, published catalog and cannot act on a partial map.

What should be reported. A macro-averaged figure computed over a pruned catalog is uninterpretable on its own, and the fix is neither expensive nor novel. Report the number of classes the average is taken over, the coverage the product delivers, and the control of Sect. 3.6. Three numbers, one of which costs a single line of code. Without them, a paper that raises macro F1 by a quarter of a point after retiring three rare classes is reporting arithmetic, and a reader cannot tell it from a paper that improved a model.

On the criterion practitioners use. The finding that retirement by support is both the weakest criterion and non-monotone deserves emphasis, because support is the criterion that gets used. It is chosen for a defensible reason — a class with few samples is a class whose estimate is unstable — but instability of the estimate and poor separability are different properties, and support only tracks the first. A class with 1 718 parcels that the model separates cleanly contributes more to a delivered map than two abundant classes the model cannot tell apart. Choosing by measured per-class quality, on blocks disjoint from those that will be evaluated, costs nothing extra and is strictly better in every comparison we ran.

On measuring meta-learners. The three figures for the same stacking model — 0.7470, 0.6789 and 0.5360 — are not three plausible estimates to choose among. The first is contaminated, and Wolpert’s original formulation of stacking already required the leakage-free partitions it violates [23]. The third is leakage-free but penalizes the model for an aggregation artifact: averaging per-block macro scores in an imbalanced setting scores blocks over classes their parcels do not contain. Pooling the out-of-fold posteriors and scoring once, which gives 0.6789, is the estimator that answers the question actually being asked. The gap between the first and the last is wide enough to reverse a conclusion about whether the ensemble contributes anything, which is why the regime has to be stated whenever such a number is printed.

6 Limitations and Conclusion

Limitations. Both benchmarks are French, so the replication varies year, region, class distribution and model family, but not the agricultural system; whether the decomposition holds in a smallholder landscape is open. On the replication benchmark the blocks are the two administrative regions, because the published index carries no per-parcel coordinates, so our resampling captures variation within regions and not between them; we report the per-block differences for this reason. That benchmark also required imputing 1.62 % of feature cells, concentrated in ratio indices whose denominator involves a band the collection encodes as zero for masked pixels; the imputation uses the training region’s median only, and it cannot favor one mechanism over another because all mechanisms read the same posterior. Finally, our out-of-fold posteriors on the primary benchmark exist for a single fold, which is why the meta-learner of Sect. 4.5 is trained on little data; that is a limitation of the artifacts available to us, not a claim about stacking in general.

Conclusion. Macro-averaged F1 over a pruned class catalog mixes two things a reader cannot separate: what the method did, and what the smaller denominator did. We gave a protocol that separates them, and the answer on two public benchmarks is that the denominator dominates — 88 % on the first, 95 % on the second at the pre-registered operating point, and 100 % over the range where coverage stays complete and every mechanism returns an identical number. At equal coverage we find no evidence that shrinking the legend beats abstaining, and significant evidence of the reverse on the more imbalanced benchmark; the criterion practitioners report using, retirement by low support, is the weakest we tested and degrades as more classes are retired. None of this requires a new model. It requires reporting the class count, the coverage, and one control line alongside every macro-averaged figure computed over a catalog someone chose.

A Reproducibility

Every figure printed in this paper is re-derived from a file under version control whose MD5, size and producing script are recorded in a custody ledger, and a gate recomputes those digests and fails when any of them changes without being registered. The ledger currently holds 62 sealed artifacts. The two benchmarks are public. The operating points, the class universes and the principal criterion $K = \lceil C/2 \rceil$ were fixed in a written pre-registration before the replication experiment was run; the two deviations forced by the data — the choice of blocks and the treatment of non-finite feature cells — were written down as dated amendments before any contrast was computed, together with the measurement error in our own cost estimate that one of them corrected.

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